

Article

Laboratory Experiments on Passive Thermal Control of Space Habitats Using Phase-Change Materials

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Abstract: Here, we investigate the performance of phase-change materials (PCMs) in the passive thermal control of space habitats. PCMs are able to absorb and release large amounts energy in the form of latent heat during their (typically, solid-to-liquid) phase transition, which makes them an ideal choice for passive temperature control. In this study, a conceptual design of an igloo-shaped habitat is proposed. A scaled model for laboratory experiments is manufactured via 3D printing, using tap water as the PCM. The setup is used to conduct experiments and analyze PCM performance, based on temperature measurements inside and outside the habitat. Results demonstrate the effectiveness of PCMs in increasing thermal inertia and stabilizing the habitat interior temperature around the melting temperature, confirming that PCMs can be a suitable alternative for passive thermal control. The present study holds significant interest for the future of space exploration, with the emerging need to design habitats that are capable of accommodating astronauts.

Keywords: space habitats; phase-change materials; experiments



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1. Introduction

In recent years, there has been a notable increase in studies and applications related to the utilization of phase-change materials (PCMs) for thermal control and energy storage [1]. PCMs possess the distinctive capacity to absorb and release large amounts of energy at a nearly constant temperature in the form of latent heat. In other words, when a PCM is integrated into a system as a thermal control device, it absorbs latent heat as the system heats up and releases it as the system cools down, thereby enhancing thermal inertia and maintaining the system close to the phase-change temperature (T_M).

This simple principle offers significant potential for innovative applications in both passive and active control systems, spanning various technological domains, such as manufacturing [2], food storage [3], net-zero energy buildings (NZEBs) [4,5], renewable energy [6], electronics [7], and space exploration [8,9]. A wide range of organic and inorganic PCMs are available nowadays [10], each with specific properties that make them suitable for such variety of applications.

Despite their low thermal conductivity, n-alkanes are one class of organic PCMs that has become popular because of their moderate T_M , compatible with habitability, and good chemical stability and compatibility with most materials [11]. These characteristics, in particular, make them attractive for space missions, where systems can experience significant temperature fluctuations resulting from the cyclical exposure to solar radiation or dissipated heat [12], and maintenance activities are generally complicated, with an additional concern related to life-limited items.

Nonetheless, there is a long history of PCM use in space exploration, going back to the Apollo 15 Lunar Rover Vehicle or the Skylab SL-1 missions, among others [13]. Future initiatives are also expected to benefit from the capability of PCMs to provide simple,

passive, and efficient thermal control. The ESA Moon Village program [14], for example, will likely require managing the extreme temperature differences between the lunar day and night to accommodate astronauts in long-duration missions. In this context, the main focus of major space agencies is to explore the potential for extraterrestrial life, and NASA has stated an ambitious goal of sending astronauts to Mars between the late 2030s and early 2040s. This new era will definitely require the development of space habitats that are capable of accommodating humans in challenging and inhospitable extraterrestrial environments.

In response to this, scientists and engineers are actively engaged in conceiving conceptual designs of Martian and lunar habitats. As an example, researchers have recently presented a human base to support long-term missions on Mars, comprising a central structure, a landing module, and an inflatable dome-shaped membrane that can be unfurled; see Figure 1a. The primary objective of this design is to create a well-isolated and protective interior zone with favorable pressure and temperature conditions, effectively shielding astronauts from the harsh Martian surroundings [15].



Figure 1. Conceptual design of Martian habitats: (a) central structure with an inflatable dome-shaped membrane, (b) 3D-printed igloo-shaped habitats. Adapted from Refs. [15,16].

Besides these conceptual designs, the way habitats will be constructed is also a topic of current interest. Among different alternatives, the potential utilization of 3D-printing technology offers numerous advantages [17]. From an application perspective, 3D printing is presently being used from early design to mass production stages in the aerospace, medical, automotive, jewelry, and architecture industries, in addition to many other areas [18], supporting the manufacturing of components with plastics, metals, ceramics, resins, and other materials. In the context of space exploration, major space agencies have plans for lunar and Martian missions where 3D printing is expected to play a crucial role in constructing bases and habitats for astronauts [16]. These projects involve on-site manufacturing of components and structures using local materials, such as lunar regolith, as raw material for printing [17]. The idea is to harness materials directly from the Martian or lunar soils, thereby significantly reducing the costs associated with the transport of large payloads and construction materials from Earth; this concept is known as *in-situ resource utilization* (ISRU) [19]. Related efforts have resulted in several noteworthy proposals; Figure 1b illustrates some of them, as documented in Ref. [16], including the winning entry from the 3D design competition organized by NASA.

These cutting-edge ideas have already pushed the boundaries of habitat design and could pave the way for sustainable human presence on Mars or the Moon in the future. Nowadays, manned missions are generally equipped with complex cooling loops and other active system like *louvres*, which help modulate and control the amount of heat absorbed and emitted via radiation. Despite being effective, this type of active systems entails a certain level of energy consumption and periodic maintenance activities, and they are more sensitive to degradation over time. In this context, PCMs can be a promising alternative

for passive thermal control, integrated in these habitats to passively maintain a suitable temperature in the interior, reducing maintenance and associated costs.

On Earth, there are several examples of PCM use for thermal conditioning in buildings. NZEBs, for instance, take advantage of PCMs and integrate them in trombe walls, building blocks, ceiling boards, or wallboards to balance the energy budget of the building [4,20,21]. In terms of energy consumption, NZEBs are a representative example of PCM-based enhancement of efficiency, where they supplement the usual thermal control implemented in active buildings and reduce the consumption associated with heating and cooling devices that usually represents a significant part of the building total. In space habitats, however, the use of PCMs for temperature control was only analyzed numerically very recently in Ref. [12], and there has been no extensive research on the topic that we are aware of. These authors suggest the feasibility of such PCM-based thermal regulation of the habitat, providing an estimate of the quantity of PCMs needed to ensure effective control and basic design guidelines to select the painting of the external habitat wall.

In this work, we extend this analysis and introduce a more detailed description of the habitat concept and its design. Then, a scaled-up laboratory experiment is developed using 3D printing to evaluate the use of PCMs for such passive temperature regulation via ground experiments; no previous research has attempted such type of experimental study. The manuscript is organized as follows. In Section 2, the space habitat design is introduced. The experiment setup is then presented in Section 3. Results from melting, solidification and cyclic experiments are discussed in Section 4. Conclusions and lines for future work are offered in Section 5. To the best of our knowledge, this manuscript explores, for the first time, the conceptual design of a PCM-based thermally controlled habitat by means of scaled-up laboratory experiments, aiming to evaluate PCM performance and the associated thermal behavior of the system.

2. Conceptual Design of the Space Habitat

As discussed above, the increased interest in lunar and Martian expeditions has opened a wide range of opportunities for the design of space habitats that align with the needs of such long-term missions. Following recent work [12], we propose a conceptual design of an igloo-shaped habitat made of regolith and PCM. The structure can be constructed either via regolith blocks filled with PCM, or via 3D printing using two co-axial nozzles that extrude regolith [17,22,23] and the PCM. The first block-based concept is shown in the upper sketch of Figure 2.

Within the scope of this manuscript, we draw attention to the co-axially 3D-printed alternative. This design can be virtually separated into two concentric semi-spherical caps of regolith, leaving an empty volume in between, which serves as a reservoir for the PCM. During repeated cycles of exposure to solar radiation and eclipses, the PCM would undergo periodic melting and solidification, stabilizing the interior temperature of the habitat around T_M . This design relies on the PCM's ability to increase thermal inertia and requires an adequate balance of heat fluxes at the external habitat boundaries (solar radiation, radiating emissions, convective cooling, etc.) and its internal energy dissipation (life support systems, crew, etc.).

As mentioned above, this concept was recently explored in Ref. [12], where PCM performance was investigated numerically considering only thermal conduction, a scenario with practical application to microgravity or reduced-gravity environments. The internal habitat temperature was analyzed for a wide range of governing parameters, including the following: the selected PCM and its quantity, characterized by its physical properties and the habitat wall thickness (L), respectively; the thermo-optical properties—absorptivity (α) and emissivity (ϵ)—at the external boundary exposed to solar radiation; and the fraction of illumination compared to the cycle period (τ_i). It was found that the thermo-optical properties at the external radiated boundary, characterized by the absorptivity–emissivity ratio (α/ϵ), play a key role in the system response and largely define the optimal design of the habitat. The optimal selection,

$$\left(\frac{\alpha}{\varepsilon}\right)_{\text{opt}} = \frac{\sigma}{\phi_S} T_M^4 \tau_i^{-1}, \quad (1)$$

balances the heat absorbed by the PCM during illumination, and the heat released during eclipse. Here, $\sigma \simeq 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is the Stefan–Boltzmann constant and ϕ_S refers to the average solar heat flux over the illumination fraction of the solar radiation cycle. This working principle is illustrated in the lower sketch of Figure 2, where the interior habitat temperature is maintained at T_M . In practice, the thermo-optical properties of regolith will not be optimal from a design perspective and the use of an appropriate painting or coating on the external habitat wall would be required [24].

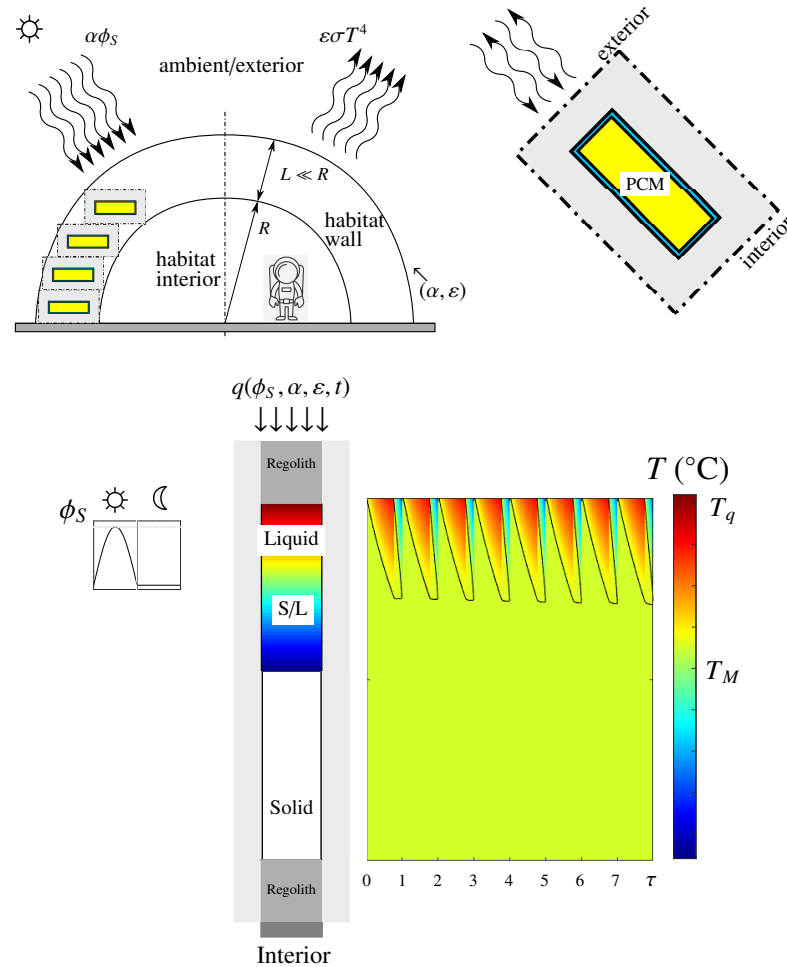


Figure 2. Conceptual design of the habitat: a 3D-printed semi-spherical habitat made of two concentric regolith caps filled with PCM. As sketched below, the PCM undergoes melting/solidification cycles during the periodic exposure to solar radiation and eclipses, and acts to stabilize the interior habitat temperature around T_M . The optimal design, where the heat absorbed/released by the PCM during repeated illumination/eclipse cycles is balanced, requires an adequate selection of thermo-optical properties at the external habitat wall. Adapted from Ref. [12].

In this work, we further estimate the dimensions of the habitat using as reference the Columbus module of the International Space Station (ISS). Columbus features a cylindrical section with a radius of 2.1 m and is designed to accommodate up to three astronauts, providing a dedicated workspace for scientific activities [25]. Similarly, our habitat is projected to accommodate two individuals and provide them with protection against the environmental conditions they would be exposed to: the regolith caps would assure structural integrity, while the PCM is expected to be continuously cycling between solid and liquid, controlling the interior thermal environment in a passive manner.

The selected habitat dimensions comprise internal and external radii of 2.78 m and 3 m, respectively, with a total wall thickness of 220 mm. This structure is formed by two concentric, 60 mm thick, semi-spherical caps of regolith, and a 100 mm thick PCM volume in between. These dimensions are specifically chosen to have similar dimensions in the nozzles, facilitating the printing process [22], and providing enough structural integrity to withstand pressure changes related to the volume variation of the PCM during phase change.

Regarding the alternatives of PCMs available for use, there is a wide range of options. We emphasize that the decisive factor lies in the value of T_M of these materials, as their efficacy is based on their ability to store and release large amounts of energy during the phase-change process around that temperature. Another important factor is the specific latent heat, which determines the amount of energy that can be stored/released per unit mass. As introduced above, the (family of) organic PCMs n-alkanes display moderate T_M and are chemically stable and non-reactive. In particular, the compounds n-octadecane, n-heptadecane, and n-hexadecane have values of $T_M = 28, 21, 18^\circ\text{C}$, respectively, similar to the mean temperature of the ISS and compatible with habitable conditions. They can be thus regarded as potential PCM candidates; their thermo-physical properties are outlined in Table 1.

N-heptadecane has the most suitable value of T_M , but has a reduced value of c_L that may discard its selection, since a larger PCM volume (by a factor of roughly 50%) would be needed to provide a storage capacity similar to that of n-hexadecane or n-octadecane. N-hexadecane, in particular, can be thought of as an ideal candidate, since its slightly lower value of T_M can somewhat compensate for the supplemental contributions of planetary radiation, albedo, and different sources of internal heat dissipation (crew, life support systems, etc.) that the habitat would experience; these additional contributions were neglected in Ref. [12]. As anticipated above, note their low thermal conductivity, which may compromise performance. The idea of exploiting thermocapillary flows to enhance heat transport in this type of PMCs has recently received lot of attention from the microgravity community; see [26–31] and references therein.

Table 1. Thermo-physical properties of n-octadecane (nC18), n-heptadecane (nC17), and n-hexadecane (nC16). Reproduced from Ref. [32].

Property	nC18	nC17	nC16
Melting temperature, T_M ($^\circ\text{C}$)	28	21	18
Latent heat, c_L (kJ kg^{-1})	243	165	237
Liquid density, ρ_l (kg m^{-3})	780	772	765
Solid density, ρ_s (kg m^{-3})	865	772	765
Liquid heat capacity, c_{pl} ($\text{J kg}^{-1} \text{K}^{-1}$)	2196	2300	2220
Solid heat capacity, c_{ps} ($\text{J kg}^{-1} \text{K}^{-1}$)	1934	1840	1950
Liquid thermal conductivity, k_l ($\text{W m}^{-1} \text{K}^{-1}$)	0.148	0.146	0.145
Solid thermal conductivity, k_s ($\text{W m}^{-1} \text{K}^{-1}$)	0.358	0.200	0.330

With this design, 3325 and 2979 L of regolith would be required to construct the external and internal semi-spherical caps, and approximately 5250 L of PCM would be needed to fill the structure, leading to an estimated total habitat mass of 13,500 kg. Considering the obvious barriers to perform in-situ tests of this concept, we propose a 1:40 scaled laboratory experiment that helps evaluate the habitat concept and PCM performance from a thermal control perspective. The experiment setup is described hereafter.

3. Experiment Setup

The heart of the experiment consists of two concentric semi-spherical caps, 3D printed from polylactic acid (PLA) with internal radii of 69.5 mm and 73.5 mm, respectively, and thicknesses of 1.5 mm. Both caps are mounted on a PLA base that, via the use of dedicated

foot prints, assures their relative positioning and leaves an evenly distributed, 2.5 mm thick space in between to be filled with the PCM.

From a thermal perspective, this scaling has various effects that can be discussed as follows. First, the scaling itself can be factored out by looking at the thermal diffusion timescale $\tau = l_c^2/\alpha$, where l_c refers to a characteristic length and α is the thermal diffusivity of the PCM. Put in other words, the results presented here for this scaled setup can be used to anticipate the dynamics for other habitat dimensions by looking at the square ratio of the associated l_c . Second, the 2.5 mm thickness between the PLA caps results in negligible convective motion during PCM melting, as measured by the Rayleigh number:

$$\text{Ra} = \frac{\rho_l g \beta \Delta T l_c^3}{\mu \alpha}, \quad (2)$$

where β and μ are the thermal expansion coefficient and dynamic viscosity of the liquid PCM, g refers to gravity, and ΔT refers to the temperature difference existing in the habitat. Note that these 1:40 scaled-up experiments are representative of the dynamics expected for the actual habitat dimensions under a reduced-gravity environment of $40^{-3} g$'s $\sim \mathcal{O}(10^{-4} g)$, i.e., under almost ideal microgravity conditions.

From an structural point of view, on the other hand, the relative dimensions between the habitat elements are preserved, and so are the expected loads. Regarding this, the difference between the internal habitat pressure and the environment is not analyzed here. However, as long as the printing conditions are selected well, so that the wall porosity is low (and controlled), the expected elastic modulus of the wall would be on the order of 200 MPa; thus, the associated deformation is negligible [33]. In addition, the reduced gravity of the typical operating space environments significantly alleviates the structural demands of counteracting the gravity field.

For the present experiments, we use tap water as a PCM, so that the interior habitat's temperature is expected to be maintained close to its melting temperature $T_M \in [0 - 4]^\circ\text{C}$. At its symmetry axis, the external cap has a hole for filling the habitat. A dedicated syringe is used to fill the igloo (i.e., the space between the two PLA caps) with 74.4 mL of water, leaving a 10% margin to compensate for the volume change occurring during the phase change. The igloo structure and the PLA base are mounted over a 20 mm thick wooden base and secured via four M4 screws. This wooden base reinforces the thermal insulation of the igloo from its bottom boundary. A CAD view of the setup is illustrated in Figure 3. As discussed above, if this setup was to be tested under radiative conditions, an adequate paint/coating should be applied to the external cap to control the heat absorption and emission via radiation at this boundary.

The igloo has an internal volume of approximately 700 mL of air at atmospheric pressure. At its center of mass, located 26 mm up from the PLA base, a type J thermocouple is installed to measure its internal temperature. This thermocouple is composed of iron–constantan (copper–nickel) materials, offering a wide measurement range of up to 760°C . Another type J thermocouple is placed out of the habitat to measure the external temperature.

Both thermocouples are calibrated using the thermal bath DC30 and a Fluke 52 II thermometer that provides a reference temperature measurement, and they are connected to a TC-08 OMEGA data acquisition module, which is able to work in a temperature range between -270 and 1820°C . The TC-08 exhibits an accuracy of 0.5°C and a resolution exceeding 0.1°C . The module is connected via USB to a laptop, where temperatures are stored in high resolution (20 bits) at a rate up to 10 Hz, and is equipped with eight channels so that up to 8 thermocouples can be connected simultaneously to perform synchronous measurements. Additionally, the manufacturer provides a basic tool to display the recorded data on a spreadsheet and/or graphical window.

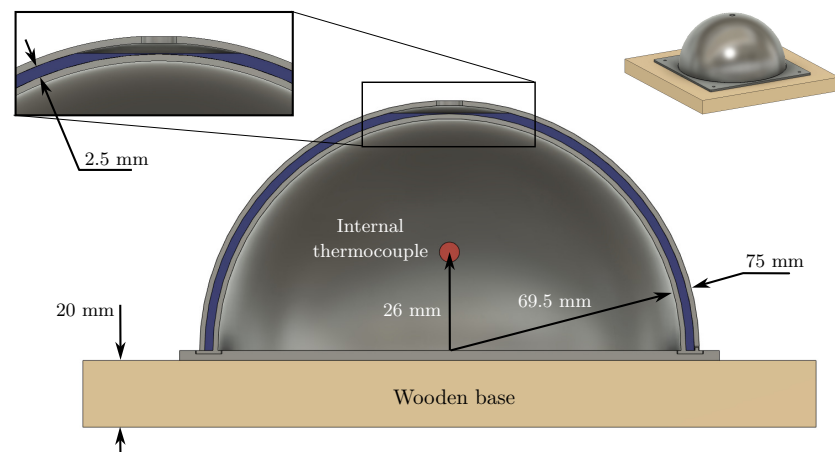


Figure 3. CAD view of the laboratory experiment setup. The principal dimensions and elements are included: 3D-printed semi-spherical caps made of PLA filled with tap water configuring the igloo, internal thermocouple, and wooden base.

To execute the experiments, the complete setup was integrated into a commercial freezer or a self-manufactured glovebox [34], which were used as environmental chambers. In both configurations, the igloo can be subjected to a controlled temperature environment at its external boundary, emulating periods of eclipse (cooling) and illumination (heating) in which the PCM solidifies or melts, respectively. The glovebox, in particular, is a partially sealed environment with transparent walls that can be heated up to 45 °C and provides a work volume of $746 \times 365 \times 536 \text{ mm}^3$. It has a front door that includes two small openings, so that the system can be operated while minimizing heat losses. The glovebox temperature is regulated via a temperature controller and a heating resistance system with an integrated fan. An image of the experimental setup installed inside the glovebox is included in Figure 4.



Figure 4. Laboratory setup installed inside the glovebox, providing a temperature-controlled environment.

Compared to the radiative condition expected during space operations, which establishes the heat flux balance, normal to the external wall boundary,

$$k\nabla T \cdot \mathbf{n} = \varepsilon\sigma T^4 - \alpha\phi_S, \quad (3)$$

imposing a controlled ambient temperature T_∞ mimics a similar scenario with a decreasing flux as the external wall temperature of the habitat evolves toward T_∞ :

$$k\nabla T \cdot \mathbf{n} = h(T - T_\infty). \quad (4)$$

Here, k refers to the thermal conductivity and $\mathbf{n}$ refers to the boundary normal vector.

Mathematically, since temperature fluctuations are $\mathcal{O}(\Delta T) = \mathcal{O}(|T_\infty - T_M|)$ and small compared to T_M , one can write $T = T_M + \hat{T}$ and linearize (3) as follows:

$$k\nabla \hat{T} \cdot \mathbf{n} = \varepsilon\sigma T_M^4 + 4\varepsilon\sigma T_M^3 \hat{T} - \alpha\phi_S = 4\varepsilon\sigma T_M^3 \left(\hat{T} + \frac{T_M}{4} - \frac{\alpha\phi_S}{4\varepsilon\sigma T_M^3} \right). \quad (5)$$

Note that this boundary problem is analogous to (4), with the substitutions:

$$h \equiv 4\varepsilon\sigma T_M^3, \quad T_\infty \equiv \frac{\alpha\phi_S}{4\varepsilon\sigma T_M^3} - \frac{T_M}{4}, \quad (6)$$

justifying this simplification.

Considering $T_M \simeq 0^\circ\text{C}$, the melting point of water, and a heat transfer coefficient $h \simeq 1 \text{ W m}^{-2} \text{ K}^{-1}$, typical of free convection of gases, one can estimate $\varepsilon \simeq 0.2167$. Furthermore, considering $T_\infty \simeq 40^\circ\text{C}$, given the thermal envelope of the glovebox [34], and $\phi_S = 1367 \text{ W m}^{-2}$, the mean solar flux experienced by spacecraft in low Earth orbit is $\alpha \simeq 0.2789$. Therefore, the present experiments would represent an optimal selection of thermo-optical properties $(\alpha/\varepsilon)_{\text{opt}} \simeq 1.287$ for $\tau_i \approx 0.2$, as anticipated by Equation (1).

In line with this, and given the used thermal chambers and envelopes, the use of water instead of n-alkanes allows for an increase of the achievable ΔT in experiments. Furthermore, the use of water does not prevent an extrapolation of the present results to those that would be obtained for n-alkanes; this fact can be justified when looking at the Stefan number:

$$\text{Ste} = \frac{c_{pl}\Delta T}{c_L}, \quad (7)$$

which governs the phase-change evolution [27]. Water, which has values of $c_{pl} \simeq 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ and $c_L \simeq 330 \text{ kJ kg}^{-1}$, displays a ratio of $c_{pl}/c_L \sim \mathcal{O}(10^{-2})$, of the same order of that of n-octadecane and n-hexadecane; only a small adjustment of ΔT would be required to obtain equal Ste.

Three different types of experiments were explored: solidification, melting, and cyclic experiments. To execute solidification experiments, the setup, initially a room temperature, was placed inside the freezer and kept there until the entire PCM volume was frozen and both thermocouples showed steady readings at approximately the same temperature below T_M . The primary objective of this test was to estimate the required time to solidify the total PCM volume, and to check the correct functioning of the temperature acquisitions.

Once the solidification was complete, the igloo can be either left at room temperature or transferred to the glovebox to execute one melting test. In either case, the setup, initially below T_M , was heated up and kept there until the entire PCM volume melted, where both thermocouples showed steady readings at approximately the same temperature. Again, the primary purpose of this test was to determine how the internal habitat temperature evolves during the melting process of the PCM and to estimate the time required to completely melt the PCM under (nearly) controlled external conditions.

For statistical and uncertainty analysis purposes, these experiments were repeated at least 20 times to check the repeatability of the results. For melting experiments, in particular,

different settings of the external temperatures were explored: tests in the glovebox were conducted at 35 and 40 °C, and an additional set of tests were conducted in which the setup was left at laboratory conditions around 25 °C.

Finally, we explored cyclic experiments, with which we aimed to mimic more realistic operational conditions, where the space habitat is exposed to repeated cycles of illumination and eclipse. This type of experiments was conducted with the aim of evaluating the PCM's effectiveness in stabilizing the internal temperature of the igloo around T_M over repeated thermal cycles, where the PCM undergoes periodic melting and solidification phases. The setup was subjected to alternating thermal environments in the freezer and the lab. The complete sequence of cycles consisted of five solidification–melting periods, with a continuous acquisition of temperature measurements. The duration of each period was estimated based on the average melting and solidification times obtained with previous experiments. For comparison purposes, two cyclic experiments were conducted: one with the water-filled igloo, and the other without water for reference.

In close connection with more realistic operational conditions, the habitat's exposure to vacuum or low pressure was expected to primarily influence the structural loads and habitat integrity, as discussed above. Therefore, the present experiments were performed at an ambient pressure. Future analyses may include the use of a thermally controlled vacuum chamber equipped with a sun simulator, like those used for spacecraft testing, qualification, and acceptance. Such experiments would reconfirm the present results under more extreme environmental conditions, but, at this early research stage, the use of such facility is not affordable.

A summary and discussion of the results is presented hereafter.

4. Results

Figure 5 illustrates the results obtained in solidification (a) and melting (b) experiments. Note that, in both cases, the temperature inside the habitat (T_{hab}) and the ambient temperature (T_∞) are represented. One can easily observe that T_{hab} remains nearly constant close to 0–4 °C during a certain lapse of the experiment, where the phase transition of water occurs.

Focusing first on the solidification test shown in panel (a), a decrease in T_{hab} was observed when the igloo was placed inside the freezer, followed by a phase where it remained approximately constant, near 0 °C. During this period, the phase change took place. Once the PCM was fully solidified, the temperature decreased again until it reached approximately T_∞ . The solidification time was obtained from the period in which the temperature remained constant. A short sub-cooling lapse was observed. Note that the setup included a small opening at the symmetry axis for filling. Along this thermal path, the effective conductivity (insulation) was larger (worse) and allowed for a faster local decrease in the interior habitat temperature. Since the interior air was warmer, natural convection acted by pulling down the cold air beneath the opening; this fact is signaled in the thermocouple readings. As this convective flow developed, it helped to homogenize the interior temperature of the habitat to a value slightly below 0 °C.

During the subsequent melting experiment [panel (b)], we can observe the opposite process. The interior temperature quickly rose as the igloo exited the freezer and was placed inside the glovebox, until it reached a value close to T_M . At this point, a significant amount of heat was absorbed, without a noticeable increase in T_{hab} , effectively storing latent heat. Once all the material was melted, the temperature increased again to match T_∞ .

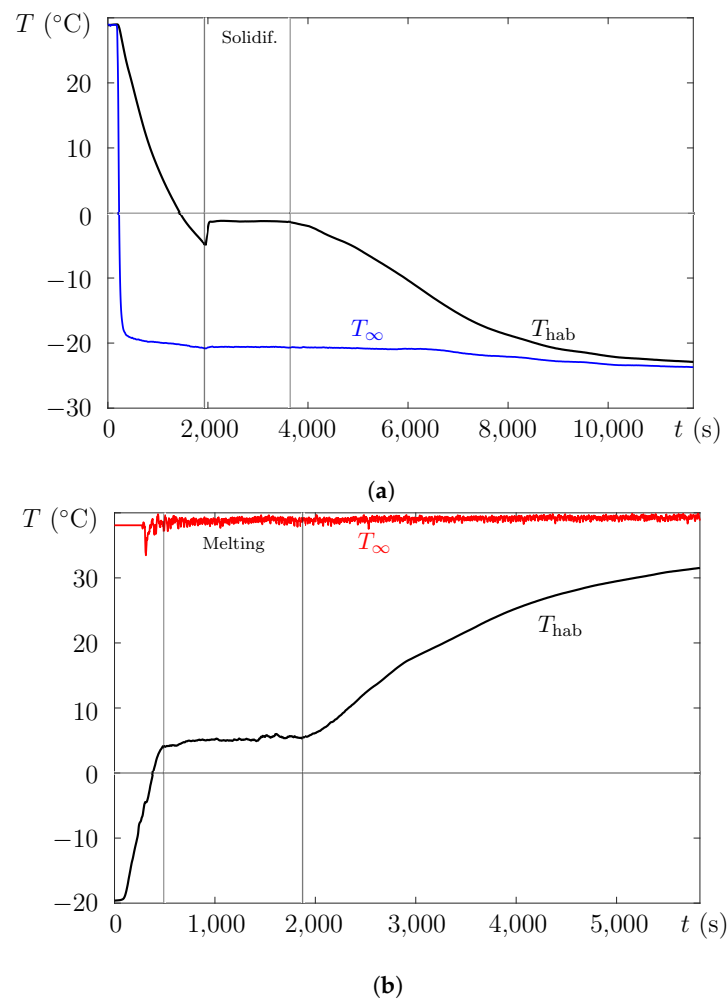


Figure 5. Time evolution of the internal (T_{hab}) and external (T_{∞}) temperatures in (a) the solidification and (b) melting experiments. The associated solidification and melting periods are labeled, characterized by a nearly constant value of T_{hab} close to 0–4 $^{\circ}\text{C}$.

Ideally, one would expect T_{hab} to remain constant at 0–4 $^{\circ}\text{C}$. However, there is a certain amount of heat introduced from the external environment. In line with the above discussion on the sub-cooling lapse, the fact that a faster increase in air temperature beneath the filling opening was gravitationally stable precludes the development of convective motion within the habitat. The thermocouple, however, measures a slightly larger temperature than 0–4 $^{\circ}\text{C}$, given the worse thermal insulation at the habitat symmetry axis.

After 20 repetitions of these tests, solidification and melting times can be estimated—focusing on the period during which the temperature remains nearly constant—and analyzed statistically. Solidification experiments provide an average freezing time of 30 min with an standard deviation of 7 min. Melting experiments result in average melting times of 25 min 32 s and 22 min 40 s with deviations of 2 min 45 s and 1 min for $T_{\infty} = 35, 40^{\circ}\text{C}$. In these experiments, the ambient temperature fluctuates $\pm 0.5^{\circ}\text{C}$ around its mean value.

Overall, these experiments evidence the effectiveness and functionality of the setup, as well as the appropriate performance of the PCM. This efficacy is reflected in the presence of an interval where T_{hab} remains nearly constant and close to T_M during its phase transition; the PCM is able to passively maintain T_{hab} . The duration of solidification and melting intervals are similar in all the experiments executed.

Finally, results from cyclic experiments are shown in Figure 6. The zones with decreasing T_{∞} correspond to the solidification periods, while those with increasing T_{∞} correspond

to the melting periods. The durations of these periods were selected in accord to the melting/solidification times estimated above.

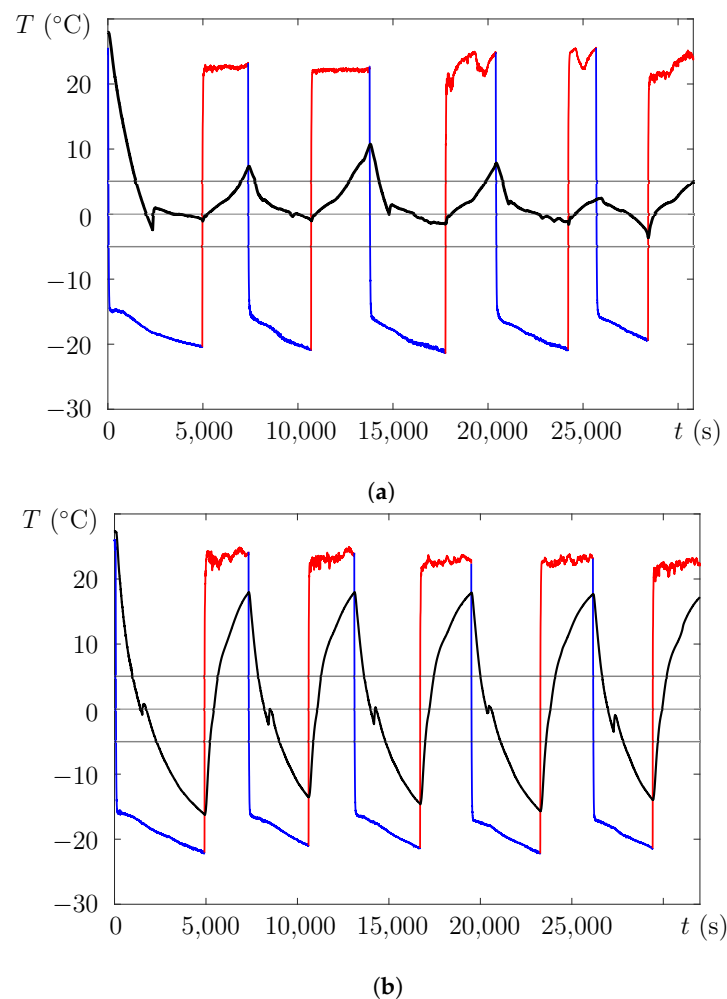


Figure 6. Time evolution of T_{hab} (black) and T_{∞} in cyclic experiments: (a) with PCM; (b) without PCM. The associated solidification and melting periods are colored in blue and red, respectively.

When examining the PCM-filled test, a pattern is observed that repeats over time, confirming a consistent thermal behavior and control of the PCM. It is observed that the temperature at the start of each solidification is the same as the temperature at the end of each fusion, which corresponds to the inflection point during fusion. The horizontal lines in the graph indicate the minimum and maximum values $T_{hab} \simeq \pm 5$ °C measured in melting/solidification experiments, marking the temperature interval where one would expect fluctuations of the internal habitat temperature. Consistent with previous experiments, the thermal response of the system also displays short sub-cooling phases.

A pattern can also be observed in the test without PCM. The igloo maintains a constant exposure both inside and outside the freezer for similar periods of time, and larger fluctuations in T_{hab} are measured, including sub-cooling. Indeed, if one overlays both scenarios, it can be observed that the temperature is significantly more stabilized in the presence of the PCM during the same time period. This phenomenon demonstrates the PCM's ability to regulate and maintain the interior habitat temperature close to T_M .

In short, these experiments successfully demonstrated that it is feasible to exert effective thermal control within the interior of a space habitat that is subjected to cyclic environmental changes, emulating repeated cycles of illumination and eclipse.

5. Conclusions and Future Work

The ambition of leading space agencies to carry out long-term missions to Mars and the Moon has emphasized the need to come up with designs of space habitats that would be capable of protecting astronauts from the extreme environmental conditions they could face. This manuscript focused on the conceptual design of a space habitat that integrates a PCM for its passive thermal control. PCMs are characterized by their ability to efficiently absorb and release energy during phase transitions while maintaining a (nearly-) constant temperature, a fact that makes them attractive for temperature control.

We proposed an igloo-shaped habitat, comprising two concentric semi-spherical caps of regolith that hold a PCM in between. During repeated cycles of exposure to solar radiation and eclipses, the PCM would undergo cyclic melting and solidification and stabilize the interior temperature of the habitat around T_M . This design relies on the PCM ability to increase thermal inertia and requires an adequate energy balance between the heat exchange occurring at the habitat boundaries and its internal energy dissipation.

A 1:40-scaled laboratory experiment was designed and manufactured to evaluate the habitat concept and PCM performance from a thermal control perspective. The heart of the experiment consisted of two concentric semi-spherical caps 3D printed from PLA, with an evenly distributed space in between that was filled with tap water, used as a PCM, with a value of $T_M \in [0 - 4]^\circ\text{C}$. The igloo was equipped with two thermocouples to measure the interior temperature T_{hab} and its external conditions T_∞ . Experiments were performed in a commercial freezer or a self-manufactured glovebox [34], used as environmental chambers. In both configurations, the igloo was subjected to controlled temperatures at its external boundary, emulating periods of eclipse (cooling) and illumination (heating) in which the PCM solidifies or melts, respectively.

Three different types of experiments were explored—solidification, melting, and cyclic experiments—validating the effectiveness and functionality of the setup and the habitat's conceptual design. During melting and solidification tests, the presence of a time interval with a (nearly-) constant $T_{\text{hab}} \sim 0^\circ\text{C}$ demonstrated the PCM's ability to passively maintain the interior temperature during its phase transition. Cyclic experiments, on the other hand, allowed us to assess the temperature's stability in the habitat by comparing its internal temperature with and without PCM. It was evident that the presence of water significantly stabilized T_{hab} around T_M , again demonstrating the PCM's ability to regulate the interior habitat's temperature in a passive manner.

Different research routes are proposed for future work. First, several improvements can be made in the setup. These include, but are not limited to, the better design of a thermal insulation system for the habitat floor, the integration and testing of various habitats in parallel with different characteristics (i.e., different filling ratios, geometries, printing materials, etc.) to allow for direct comparison of the internal environments under (nearly-) identical external conditions, and the use of a commercial environmental chamber that allows for controlled (and programmable) temperature conditions. In line with this, one can perform experiments in solar simulators of commercial use to better reproduce the expected operational conditions, including the exposure to low pressure or vacuum. Considering different materials, one may evaluate the possibility of using simulants of surface materials, while any environmental chamber should allow the analysis of other PCMs, like n-octadecane, n-heptadecane, or n-hexadecane; these organic PCMs are known for their moderate values of T_M —compatible with habitable conditions—and long-term stability, being potential choices for practical applications. Finally, these laboratory experiments can be used to tune-up numerical models that predict PCM behavior in microgravity, and help optimize the conceptual design of the habitat; this numerical analysis will be undertaken elsewhere.

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